



Science

Stage 9

Paper 2

2025

Cambridge Lower Secondary Progression Test

Mark Scheme

General guidelines on marking

Many descriptive answers can be expressed in a variety of ways. Professional judgement can be used in these cases, providing it matches the marking points and further information in the mark scheme.

Answers may have words spelt incorrectly. Credit is normally given for phonetically correct answers, unless the word has a scientifically different meaning. For example, where the answer should be antennae, credit will be given for antenna but not for anthen (too close to anther).

Only the science is being assessed, so answers do not need to be grammatically correct.

Significant figures will be indicated in the question or in the mark scheme.

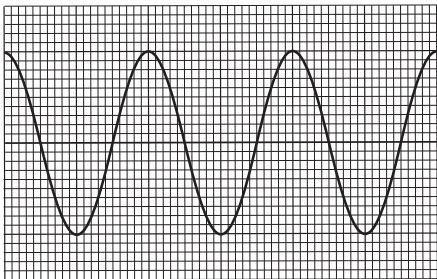
Unless specified all marking points are independent.

Annotations and abbreviations

/ or	alternate responses for the same marking point
() brackets	the words or units in brackets do not need to be stated, for example, (recycles or releases or provides) minerals = minerals scores the mark
<u>underline</u>	exact word is required
Accept	an acceptable response
Do not accept	indicates an incorrect response that would contradict another otherwise correct alternative
Ignore	indicates an irrelevant answer that is not creditworthy. Full marks can still be achieved even with answers that are ignored.
Note	provides extra information when necessary
ecf	error carried forward; marks are awarded if an incorrect response has been carried forward from earlier working provided the subsequent working is correct
ora	or reverse argument; for example, as mass increases, volume increases could be written as mass decreases, volume decreases

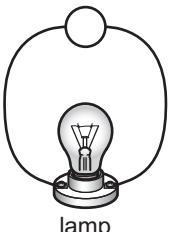
Question	Answer	Marks	Further Information
1(a)	root <u>hair</u> (cell)	1	Note root on its own is not sufficient
1(b)	absorption	1	Accept osmosis or diffusion
1(c)	xylem	1	
1(d)	transpiration	1	Accept evaporation or diffusion

Question	Answer	Marks	Further Information
2(a)(i)	covalent	1	
2(a)(ii)	loses	1	Accept gives away
2(b)	attraction/bond between a positively charged ion and a negatively charged ion	1	Accept electrostatic attraction between ions

Question	Answer	Marks	Further Information
3	 sound waveform B	1	Note drawn waveform must have the same amplitude and frequency by eye but all the peaks and troughs are opposite each other as a mirror image of waveform A
Question	Answer	Marks	Further Information
4(a)	less rain	1	more than one answer circled = 0 marks Accept any indication of the correct answer, e.g. ticking or underlining, but circling takes precedence
4(b)	any two from less food less water to drink lack of hygiene or increase risk of diseases migration of people (to find food or to find jobs) (idea of) loss of plant species or loss of animal species	2	each correct answer = 1 mark Accept increase in diseases such as cholera or typhoid Accept other suitable impacts of desertification, e.g. lack to trees to build houses

Question	Answer	Marks	Further Information
5(a)	carbon dioxide water oxygen	3	each correct answer = 1 mark Note carbon dioxide and water can be written in any order
5(b)	photosynthesis chloroplasts	2	each correct answer = 1 mark

Question	Answer	Marks	Further Information
6(a)	$\text{Zn} + \text{CuSO}_4 \longrightarrow \text{ZnSO}_4 + \text{Cu}$	1	Accept = for $\longrightarrow$ Accept reactants in either order Accept products in either order Ignore words
6(b)	no (no mark) (idea that) copper is less reactive than magnesium ora	1	yes ticked = 0 marks for the question Accept copper is below magnesium in the reactivity series ora Note copper is less reactive is not sufficient Do not accept copper is less reactive than magnesium sulfate
6(c)	yes (no mark) (idea that) iron is more reactive than silver ora	1	no ticked = 0 marks for the question Accept iron is above silver in the reactivity series ora Note iron is more reactive is not sufficient Do not accept iron is more reactive than silver nitrate

Question	Answer	Marks	Further Information
7(a)(i)	voltmeter	1	Do not accept voltameter or voltage meter
7(b)(ii)	meter drawn across the lamp, e.g.  lamp	1	Note any symbol may be used for the meter, e.g. just a circle or a picture of a meter
7(b)	(resistance =) $\frac{\text{voltage}}{\text{current}}$ or $\frac{1.5}{0.6}$ 2.5 (Ω)	2	Accept correct answer with or without working = 2 marks correct equation or substitution = 1 mark correct calculation = 1 mark Note answer on answer line takes precedence

Question	Answer	Marks	Further Information
8(a)	(idea that) to remove the threat of the asteroid hitting the Earth	1	Accept (idea that) NASA want to practice destroying asteroids
8(b)	any one from (idea of) mass extinction (idea of) shock waves (idea of) destruction of buildings or destruction of cities (idea of) dust clouds that block out light or dust blocks the Sun (idea of) increase in temperature where the asteroid hits	1	Accept many people would die Ignore just kills animals or kills humans Accept flattens buildings

Question	Answer	Marks	Further Information
9	(A) endive (B) purslane (C) mizuna (D) oak leaf (E) romaine	2	all five correct = 2 marks two, three or four correct = 1 mark one correct = 0 marks

Question	Answer	Marks	Further Information
10(a)	(density =) mass $\div$ volume or $150 \div 25$ (density =) 6 g/cm^3	3	Accept 6 g/cm^3 with or without working = 3 marks Accept 6 with or without working = 2 marks equation or correct substitution = 1 mark Accept $d = m/v$ or $m \div v$ correct units = 1 mark Note density and units on answer line takes precedence
10(b)	giant	1	Ignore metallic or ionic or molecular or covalent

Question	Answer	Marks	Further Information
11(a)	thermal energy is transferred from block A to block B	1	Accept thermal energy is dissipated Accept heat for thermal energy
11(b)	(idea that) the two blocks become the same temperature	1	Accept the block A gets cooler and the block B get hotter Accept they both contain the same amount of thermal energy Accept heat for thermal energy Ignore thermal energy transfer stops

Question	Answer	Marks	Further Information
12	(heading) time in min(utes) (heading) volume in cm ³ data correctly entered without units	3	each correct answer = 1 mark Accept time and volume in the headings without units = 1 mark Accept min(utes) and cm ³ in the headings without physical quantity= 1 mark Ignore order of the data

Question	Answer	Marks	Further Information
13	any two from liquid water becomes a gas or water becomes water vapour (idea that) thermal energy is transferred from the skin (to the water) to the air (evaporation) is endothermic or requires thermal energy	2	each correct answer = 1 mark Accept faster particles leave the liquid or particles with more energy escape the water Accept slower particles left behind with liquid water or particles with less energy left in the liquid water

Question	Answer	Marks	Further Information
14(a)	any one from (idea that) emissions are too low to show on the graph (idea that) the emissions were not measured (idea that) fossil fuels were not used then (idea that) the population was much lower in 1850 to 1865 (idea that) the only carbon dioxide released is in the slow carbon cycle	1	Ignore just no carbon dioxide is released
14(b)	(description of trend) a slow increase and then a rapid increase (explanation) (idea that) slow increase linked to just the slow carbon cycle and rapid increase linked to the (slow and) fast carbon cycle	2	each correct answer = 1 mark Accept increases
14(c)	any value between 7 and 10.5 (billions of tonnes)	1	

Question	Answer	Marks	Further Information
15(a)	(measurement) decrease in volume of red dye solution or decrease in mass of red dye solution or distance travelled by red dye solution and (equipment) measuring cylinder or a balance or ruler (measurement) time for red dye solution to reach leaf and (equipment) stop-watch	2	each measurement and linked equipment = 1 mark Accept water for red dye solution Accept scales Ignore weighing or top pan or lever arm or electronic Accept stop-clock
15(b)	more reliable or to identify anomalous results or to see if results can be reproduced	1	Do not accept for accuracy or fair testing

Question	Answer	Marks	Further Information
16(a)	(risk) hot water may scald and (control) wear heatproof gloves (risk) dye may be poisonous or dye may be harmful or dye may stain or dye may splash into eyes and (control) wear eye protection or wear goggles or wear gloves	2	each risk correctly linked to a control = 1 mark Accept hot water may cause burns Note mark answers wherever seen
16(b)	(predictions) (idea that) cold blue water sinks and hot red water floats (explanation for cold blue water) cold water is more dense than the water at room temperature (explanation for hot red water) hot water is less dense than the water at room temperature	3	each correct answer = 1 mark